Data Mining

Web, Spatial and Temporal Mining

Web Mining

- Web mining is the process of applying data mining techniques to discover patterns from web data, including web pages, server logs, and hyperlinks.
  - Web content mining
  - Web structure mining
  - Web Usage Mining

Web Mining


Web Mining
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Web Content Mining
Web Structure Mining
Web Usage Mining
Text
Image
Audio
Video
Structure Record
Hyperlink
Document Structure
Intra-Doc Hyperlink
Inter- Doc Hyperlink
Web Server logs
Application Server logs
Application Level logs

Web Content Mining

- Extracting useful information from web page contents
- Analyses the actual data on pages to find facts, product descriptions, reviews, and other information.
- Data types: text, images, audio, video
- Techniques:
  - Text mining and natural language processing (NLP)
  - Multimedia data mining
- Applications: news classification, product recommendation

Web Structure Mining

- Analyzes the link structure of a website to understand relationships between pages.
- Identifies hubs (pages with many outgoing links) and authorities (pages with many incoming links) and can use algorithms like PageRank.
- Helps improve search results by understanding how pages are connected.
- Representation using graphs (nodes = pages, edges = links)
- Algorithms:
  - HITS (Hyperlink-Induced Topic Search)
  - PageRank
- Applications: ranking pages, community detection

Web Usage Mining

- Discovering patterns from web log data
- Phases: preprocessing, pattern discovery, pattern analysis
- Techniques: association, clustering, sequential pattern mining
- Applications: personalization, user profiling, website optimization

Overview of Spatial Data

- Spatial data: data related to space and location
- Types: raster (images, grids), vector (points, lines, polygons)
- Spatial databases and GIS integration
- Challenges: high dimensionality, spatial autocorrelation

Spatial Data Mining Primitives

- Spatial objects, spatial attributes, and relationships
- Spatial predicates: intersects, adjacent, within, overlaps
- Query examples using spatial SQL
- Importance in geographic and environmental analysis

Spatial Rules

- Spatial association rules: relationships among spatial features
- Example: "*If a region has high rainfall, then it has dense vegetation.*"
- Rule types: spatial–spatial, spatial–nonspatial
- Evaluation using support and confidence

Spatial Classification Algorithms

- Goal: classify spatial objects based on spatial and non-spatial attributes
- Approaches:
  - Decision tree with spatial attributes
  - Spatial Bayesian classifier
- Example: land-use classification from satellite imagery

Spatial Clustering Algorithms

- Grouping spatial objects into meaningful clusters
- Algorithms:
  - DBSCAN (Density-Based Spatial Clustering)
  - STING (Statistical Information Grid)
  - CLARANS (Clustering Large Applications based on RANdomized Search)
- Applications: urban planning, ecology, geoscience

Modeling Temporal Events

- Temporal data: data with time-dependent behavior
- Temporal dimensions: valid time and transaction time
- Event sequences and temporal patterns
- Importance in stock, medical, and weather analysis

Time Series Analysis

- Definition: sequence of observations over time
- Tasks: trend detection, forecasting, anomaly detection
- Methods: ARIMA, exponential smoothing, neural networks
- Example: predicting temperature or sales over months

Pattern Detection in Temporal Data

- Identifying periodic patterns and trends
- Temporal autocorrelation
- Sliding window and dynamic time warping (DTW) methods

Sequence Mining

- Sequential pattern mining: discovering frequent event sequences
- Algorithms: GSP (Generalized Sequential Pattern), PrefixSpan
- Example: user navigation paths, purchase sequences

Temporal Association Rules

- Extending association rules with time dimension
- Example: *"If patient takes Drug A, within 3 days symptom B appears."*
- Measures: temporal support, temporal confidence
- Applications: healthcare, retail, finance

Summary

- Web, Spatial, and Temporal mining expand classical data mining
- Common goal: extract patterns from complex, structured data
- Future directions: spatio-temporal mining, deep learning for web data